Miinla Thesis Applicant Task

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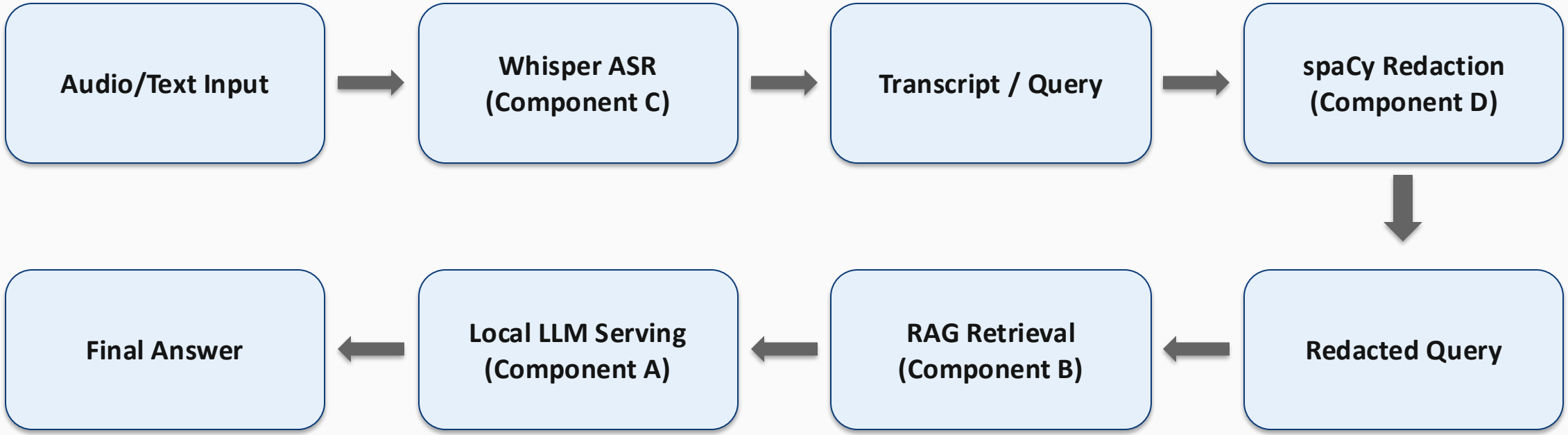
Date: 07 April 2026
Hardware: MacBook Air M1, 8 GB RAM

Implementation Overview

- Goal: To build a minimal end-to-end prototype of Miinla's local AI pipeline across LLM, RAG, ASR, and PII redaction.
- Implemented four components: Ollama + FastAPI, RAG with ChromaDB, Whisper ASR, and spaCy redaction.
- Connected the system from audio/text input to final answer through one integrated flow.
- Evaluated the prototype on five German Q&A examples, including failure cases and tradeoff analysis.

System Architecture



Component A: FastAPI + Ollama

Local LLM Serving Endpoint

- Served phi3:mini locally through a FastAPI endpoint on an M1 CPU.
- Chose phi3:mini because it is lightweight enough for an 8 GB RAM laptop setup.
- Observed roughly 13s latency, which is slower than cloud APIs but keeps inference local and data-isolated.
- Endpoint returns a structured JSON response and logs request latency.

```
(venv) bhavika@Bhavikas-MacBook-Air Miinla Task Test % python component_a/LLM_serving.py
INFO:     Started server process [39882]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

```
(venv) bhavika@Bhavikas-MacBook-Air Miinla Task Test % curl -X POST http://localhost:8000/generate \
  -H "Content-Type: application/json" \
  -d '{"prompt": "Was ist Datenschutz in einem Satz?"}'
{"answer":"Datenschutz bezieht sich auf die rechtlichen und ethischen Maßnahmen, um den Schutz personenbezogener Daten zu gewährleiste
n. Es handelt sich um ein wesentliches Prinzip im digitalen Zeitalter, das es Einzelpersonen ermöglicht, ihre Privatsphäre und persönl
ichen Informationen vor unbefugtem Zugriff oder Missbrauch zu schützen.","model":"phi3:mini","latency_ms":13111.17}%
(venv) bhavika@Bhavikas-MacBook-Air Miinla Task Test %
```

Component B: RAG Pipeline

Architecture Choices

- Used a German DSGVO sample document as the retrieval corpus.
- Applied fixed-size chunking: 500 characters with 50-character overlap.
- Embedded chunks with intfloat/multilingual-e5-small and stored them in ChromaDB.
- Main weakness: fixed-size chunking is brittle and can break German compound nouns or semantic boundaries.

```
(venv) bhavika@Bhavikas-MacBook-Air Miinla Task Test % python component_b/Rag_Pipeline.py \
  --doc data/dsgvo_sample.txt \
  --question "Was ist die DSGVO?"
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloa
ds.
Loading weights: 100%|[redacted]| 199/199 [00:00<00:00, 5219.67it/s]
BertModel LOAD REPORT from: intfloat/multilingual-e5-small
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Frage  : Was ist die DSGVO?
Antwort: Die Datenschutz-Grundverordnung (DSGVO) ist eine Verordnung der Europäischen Union, mit der die Regeln zur Verarbeitung perso
nenbezogener Daten durch die meisten Datenverarbeiter geregelt werden. Sie betont den Schutz personenbezogener Daten und legt fest, da
ss diese nur rechtmäßig, nach Treu und Glauben und in einer für die betroffene Person nachvollziehbaren Weise verarbeitet werden müsse
n.
Latenz : 18306.48 ms
(venv) bhavika@Bhavikas-MacBook-Air Miinla Task Test % python component_b/Rag_Pipeline.py \
  --doc data/dsgvo_sample.txt \
  --question "Was ist die DSGVO?"
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloa
ds.
Loading weights: 100%|[redacted]| 199/199 [00:00<00:00, 5041.41it/s]
BertModel LOAD REPORT from: intfloat/multilingual-e5-small
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Frage  : Was ist die DSGVO?
Antwort: Die Datenschutz-Grundverordnung (DSGVO) ist eine Verordnung der Europäischen Union, welche den Schutz personenbezogener Daten
 durch Unternehmen regelt und sicherstellt. Sie betont die Notwendigkeit einer rechtmäßigen, nach Treu und Glauben verarbeitenden und
für die betroffene Person nachvollziehbaren Datennutzung.
Latenz : 11573.18 ms
```

Component C: Whisper ASR

Audio Transcription:

- Used Whisper `base` for German speech transcription.
- Forced the input language to `de` to reduce misdetection on short audio clips.
- Observed low latency of about 1.5 s to 2.4 s on local execution.
- Main error: “DSGVO” was occasionally mis-transcribed as “DSG-Vero,” showing domain-acronym weakness.

```
(venv) bhavika@Bhavikas-MacBook-Air Miinla Task Test % python component_c/ASR.py --generate-sample
100%|                                                                                  | 1111/1111 [00:01<00:00, 859.36frames/

Transcript : Was sind die wichtigsten Grundsätze der Datenschutzgrundverordnung? Die DSG-Vero schützt Personenbezogene Daten in der
ropäischen Union.
Language   : de
Latency    : 1545.52 ms
```

Component D: spaCy Redaction

- Used de_core_news_sm to detect and mask PER, ORG, and LOC entities.
- Replaced entities with typed placeholders such as [PERSON_1], [ORG_1], and [LOC_1].
- Applied reverse-order offset replacement so spans remain stable during masking.
- Main weakness: titles like "Dr." remain unmasked, and domain terms like "DSGVO" can be incorrectly tagged as organizations.

```
(venv) bhavika@Bhavikas-MacBook-Air Miinla Task Test % python component_d/PII_redaction.py

  Original : Hallo, ich bin Maria Schneider von der Mustermann GmbH in Frankfurt.
  Redacted : Hallo, ich bin [PERSON_1] von der [ORG_1] in [LOC_1].
    [PERSON_1] → Maria Schneider
    [ORG_1] → Mustermann GmbH
    [LOC_1] → Frankfurt

  Original : Dr. Klaus Müller arbeitet bei Siemens AG in München.
  Redacted : Dr. [PERSON_1] arbeitet bei [ORG_1] in [LOC_1].
    [PERSON_1] → Klaus Müller
    [ORG_1] → Siemens AG
    [LOC_1] → München

  Original : Die Datenschutzbeauftragte Frau Petra Weber ist in Berlin tätig.
  Redacted : Die Datenschutzbeauftragte Frau [PERSON_1] ist in [LOC_1] tätig.
    [PERSON_1] → Petra Weber
    [LOC_1] → Berlin

  Original : Ein anonymer Nutzer stellte eine Anfrage ohne persönliche Daten.
  Redacted : Ein anonymer Nutzer stellte eine Anfrage ohne persönliche Daten.
    No PII detected.
```

End-to-End Demo

Pipeline execution:

- Pipeline can be triggered from one flow: query input → redaction → retrieval → local generation.
- It fails fast if the Ollama server is unavailable.
- File-based logging separates internal debug details from user-facing output.

```
(venv) bhavika@Bhavikas-MacBook-Air Miinla Task Test % tail -30 logs/pipeline.log
2026-04-07 17:22:19 [INFO    ] component_d.PII_redaction — redacted : Welche Strafen drohen bei Verstößen gegen die [ORG_1]?
2026-04-07 17:22:19 [INFO    ] component_d.PII_redaction —   [ORG_1] → DSGVO
2026-04-07 17:22:19 [INFO    ] __main__ — step 3 — RAG query | doc=data/dsgvo_sample.txt
2026-04-07 17:22:25 [INFO    ] component_b.Rag_Pipeline — reusing existing DB (2 chunks)
2026-04-07 17:22:25 [INFO    ] component_b.Rag_Pipeline — retrieved top-3 chunks for query: Welche Strafen drohen bei Verstößen gegen die
ORG_1]?
2026-04-07 17:22:25 [DEBUG   ] component_b.Rag_Pipeline —   [1] dist=0.1802 | .
Artikel 5 DSGVO besagt, dass Daten rechtmäßig, nach Treu und Glauben und in ei...
2026-04-07 17:22:25 [DEBUG   ] component_b.Rag_Pipeline —   [2] dist=0.1894 | Datenschutz-Grundverordnung (DSGVO)
Die Datenschutz-Grundverordnung (DSGVO) ist ...
2026-04-07 17:22:25 [DEBUG   ] component_b.Rag_Pipeline — sending prompt to LLM (phi3:mini)
2026-04-07 17:22:39 [INFO    ] component_b.Rag_Pipeline — LLM response received — latency=13377ms
2026-04-07 17:22:39 [INFO    ] __main__ — pipeline complete | latency=13377ms answer_len=400
2026-04-07 17:24:00 [INFO    ] LLM_serving — health check passed — Ollama reachable
2026-04-07 17:24:00 [INFO    ] __main__ — LLM server reachable at http://localhost:8000/health
2026-04-07 17:24:00 [INFO    ] __main__ — step 1 — ASR skipped | text query provided
2026-04-07 17:24:00 [INFO    ] __main__ — step 2 — PII redaction
2026-04-07 17:24:00 [DEBUG   ] component_d.PII_redaction — redacted 0 entities from input text
2026-04-07 17:24:00 [INFO    ] component_d.PII_redaction — original : Was ist eine Datenschutz-Folgenabschätzung?
2026-04-07 17:24:00 [INFO    ] component_d.PII_redaction — redacted : Was ist eine Datenschutz-Folgenabschätzung?
2026-04-07 17:24:00 [INFO    ] component_d.PII_redaction — no PII detected
2026-04-07 17:24:00 [INFO    ] __main__ — step 3 — RAG query | doc=data/dsgvo_sample.txt
2026-04-07 17:24:06 [INFO    ] component_b.Rag_Pipeline — reusing existing DB (2 chunks)
2026-04-07 17:24:07 [INFO    ] component_b.Rag_Pipeline — retrieved top-3 chunks for query: Was ist eine Datenschutz-Folgenabschätzung?
2026-04-07 17:24:07 [DEBUG   ] component_b.Rag_Pipeline —   [1] dist=0.1632 | .
Artikel 5 DSGVO besagt, dass Daten rechtmäßig, nach Treu und Glauben und in ei...
2026-04-07 17:24:07 [DEBUG   ] component_b.Rag_Pipeline —   [2] dist=0.1657 | Datenschutz-Grundverordnung (DSGVO)
Die Datenschutz-Grundverordnung (DSGVO) ist ...
2026-04-07 17:24:07 [DEBUG   ] component_b.Rag_Pipeline — sending prompt to LLM (phi3:mini)
2026-04-07 17:24:20 [INFO    ] component_b.Rag_Pipeline — LLM response received — latency=12636ms
2026-04-07 17:24:20 [INFO    ] __main__ — pipeline complete | latency=12636ms answer_len=346
(venv) bhavika@Bhavikas-MacBook-Air Miinla Task Test %
```

Q&A Result Matrix

Question Theme	Pipeline Result
Personenbezogene Daten	Good Answer
Datenschutz-Folgenabschätzung	Good Answer
Rechte betroffener Personen	Mostly Good
Zweck der DSGVO	Failed due to masking [ORG_1]
Strafen bei Verstößen	Incorrect / hallucinated

Key takeaway: the pipeline orchestration works, but answer quality drops quickly when the redaction layer distorts critical legal terms.

Q&A Execution Snapshots

```
(venv) bhavika@Bhavikas-MacBook-Air Miinla Task Test % python pipeline.py --text "Was ist der Zweck der DSGVO?" --doc data/dsgvo_sampl
e.txt

============================================================
  Pipeline: Audio/Text → ASR → PII Redaction → RAG → LLM
============================================================

[STEP 1] Skipping ASR — using provided text query

  Raw Query: Was ist der Zweck der DSGVO?

[STEP 2] PII Redaction

  Original : Was ist der Zweck der DSGVO?
  Redacted : Was ist der Zweck der [ORG_1]?

[STEP 3] RAG Retrieval + LLM Generation
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloa
ds.
Loading weights: 100%|┃┃┃
```

Three Biggest Limitations

1. PII over-redaction: `DSGVO` was misclassified as ORG, which damaged retrieval context.

2. Weak retrieval setup: the document set was small, and simple character chunking did not preserve logical structure well.

3. Domain adaptation errors: Whisper struggled with niche acronyms, and the LLM hallucinated details when context quality dropped.

Production Improvements

German ERP RAG

- Sentence-aware / section-aware chunking.
- Metadata filters by module, document type, and date.
- Cross-encoder reranking.
- Domain glossary for ERP acronyms and aliases.
- Document-level access control and source-aware retrieval.

GDPR Production Design

- Stronger detection beyond base spaCy.
- Private / in-VPC vector isolation.
- Pre-storage data minimization.
- Human fallback review for uncertain cases.
- Audit trails, retention policies, and RBAC.

Priorities: First 4 Weeks

- Week 1: build a stronger evaluation set using realistic German ERP queries.
- Week 2: improve retrieval with reranking and semantic splitters.
- Week 3: extend spaCy with custom domain rules and logic pipelines.
- Week 4: add observability, audit logging, and data-footprint governance.
- Conclusion: the architecture works as a proof of concept; the next step is making it reliable under production constraints.